

Rutvi Khamar

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Education

Florida Institute of Technology

M.S. in Computer Science (GPA: 3.93 / 4.00)

Aug 2023 - May 2025

Melbourne, Florida

Relevant Coursework: Applied Discrete Mathematics, Machine Learning (coded algorithms from scratch), Big Data (Spark/Hadoop/AWS), NLP, Thesis (Early breast cancer detection using Non-negative Matrix Factorization)

Adani University

B.Tech in Information and Communication Technology (GPA: 8.93 / 10.00)

Aug 2019 - May 2023

Ahmedabad, Gujarat

Relevant Coursework: Data structures & algorithms, Object oriented programming, Data Mining & BI, Linear Algebra, Probability & Statistics, Advanced Software Engineering

Technical Skills

Programming: Python, SQL, Java

Frameworks & Tools: Numpy, Pandas, PyTorch, Tensorflow, Transformers, HuggingFace, Transformers, Dask, spaCy, Prodigy, PySpark, LangChain, LangGraph, Claude Agent SDK, MLflow, Weights & Biases, scikitlearn, matplotlib, plotly, Streamlit, Cursor, Claude code, Snowflake, dbt, Dagster

Certification: Technical AI Safety – BlueDot Impact

Covered alignment failure modes (specification gaming, reward hacking), RLHF and training-time safety techniques, model evaluations, interpretability, and harm-minimization / threat modeling.

Experience

Steppingblocks | AI/ML Engineer - Melbourne, FL

Aug 2025 – Present

- Worked directly with clients to scope record-linkage requirements, then rebuilt a pipeline crashing on their large datasets by migrating to Spark and redesigning the blocking, training, and comparison logic – lifting entity-matching accuracy from ~60% to ~80%
- Designed a job-title-to-SOC-code classification pipeline combining RAG, cross-encoders, and a domain-specific embedding model fine-tuned on resume and skills data, capturing semantic relationships between related concepts that general-purpose models miss
- Built a data-driven skills taxonomy by clustering semantically similar skills with HDBSCAN over a fine-tuned embedding model, replacing manual curation with an unsupervised workflow that scales to new skills without supervision
- Engineered a 150K-skill definition-generation pipeline on Groq's batch inference API, using chain-of-thought and few-shot prompting with typed JSON output (10 skills per call) for reliable parsing – resolving cost and latency bottlenecks that made hosted APIs prohibitive and local models unstable at scale
- Built an embedding-visualization tool to empirically benchmark general vs. domain-specific models, demonstrating tighter clustering of semantically related titles and skills – used to justify and validate model selection

L3Harris Institute for Assured Information | Research Assistant - Melbourne, FL

May 2024 – July 2024

- Conducted systematic research comparing multiple entity extraction approaches (SpaCy+ClausalE, Stanza-CoreNLP, fine-tuned LLaMA2, T5-based transformers) to identify optimal methods for converting unstructured text into structured knowledge graphs, evaluating trade-offs between accuracy, computational cost, and scalability.
- Designed and executed experiments for fine-tuning LLaMA2 for structured query generation with limited compute resources, implementing prompt engineering and parameter-efficient fine-tuning strategies to enable text-to-Cypher translation for graph database querying.
- Built end-to-end GraphRAG research prototype integrating LLMs, Neo4j graph database, to explore how graph-structured knowledge enhances LLM reasoning over complex relational data.
- Documented comparative analysis across all approaches, evaluating each method's strengths and failure modes to guide future research directions.

Projects

BFCL V4 – Inspect Evals | *Agentic Evaluation, Tool-Calling, Web Search, Memory, Python* 🔗

- Implemented BFCL V4 – the agentic tool-calling benchmark – for Inspect Evals, the open-source evaluation suite built on the UK AI Safety Institute’s Inspect framework, extending coverage beyond the existing V1–V3 categories to V4’s agentic scenarios.
- Built the evaluation harness and scoring for V4’s agentic tasks, measuring how models perform multi-hop web search, read/write persistent memory, supported by a BlueDot Impact rapid grant, and also documented in a technical write-up.

SafeCircuits – Interpretability for AI Safety Classifiers | *Mechanistic Interpretability, SAEs, LlamaGuard, PyTorch* 🔗

- Built an interpretability toolkit for a production safety classifier (LlamaGuard-7B + NVIDIA Aegis adapter), exposing *why* content is flagged safe/unsafe via token attribution, layer-ablation importance, and attention analysis.
- Trained sparse autoencoders (SAEs) on the model’s residual stream to surface interpretable features, and shipped an interactive visualization app for prompt-by-prompt auditing of safety decisions.

GraphRAG for Healthcare | *Neo4j, LLMs, Cypher, Prompt Engineering, Entity Extraction* 🔗

- Built a GraphRAG pipeline on the Semantic MEDLINE Database (SemMedDB) to reduce LLM hallucinations in healthcare – constructing a Neo4j knowledge graph and enabling natural-language querying via LLM-generated Cypher with prompt engineering and entity extraction.
- Demonstrated that graph-structured retrieval captures higher-order entity relationships missed by standard RAG, improving contextual accuracy of LLM responses in medical Q&A.

FlightMate | *LangChain, ArangoDB, AQL, NetworkX, Gradio* 🔗

- Built a multi-tool agentic chatbot over a graph database of airports, flights, and airlines – dynamically routing queries across tools for shortest-path computation, live weather lookup, and airline statistics.
- Designed a hybrid tool combining AQL graph queries with NetworkX algorithms and interactive visualizations, enabling complex relational queries with real-time graph rendering.

Research Work

Early breast cancer detection with 3D ultrasound using NMF (Thesis) | *Advisor: Dr. Mitra*

- Published and presented at SPIE Medical Imaging Conference 2025 – a novel ML approach for early-stage breast cancer detection using Non-negative Matrix Factorization on 3D ultrasound data.
- Developed and optimized multiple NMF variants with custom regularization to extract diagnostic features from DCE-US imaging data, distinguishing benign from malignant lesions without contrast agents.
- Achieved 75.5% sensitivity and 78.3% specificity using only pre-injection data – matching contrast-enhanced performance while eliminating patient risk, demonstrating that smarter feature extraction can replace invasive procedures.